16

RYERSON UNIVERSITY
DEPARTMENT OF PHYSICS
PCS 125 FALL 2007 MID-TERM EXAMINATION

DURATION: 90 minutes **DATE: October 19, 2007**

[redacted]	[redacted]	[redacted]	[redacted]
Student Number	**Section**	**PRINT first name**	**PRINT family name**

Circle your instructor's name.

Beauchemin **Ford** **Kumaradas** **Toronov** **Szmidt**

INSTRUCTIONS

- This is a closed-book exam; pens, pencils, erasers, calculators and your Ryerson ID are the only allowed items on your desk. Your Ryerson photo ID must be on your desk at all times. Do not separate any sheets from the booklet at any time.
- Check that the "bubble sheet" at your desk has your name and student number on it.
- The only calculators allowed are the Sharp EL-546 and Casio FX-991.
- **PDAs, phones, and pagers must be turned off** and **out of reach.** Coats, jackets, caps and bags must be placed out of reach.
- **SECTION A: MULTIPLE-CHOICE QUESTIONS (#1—#8)**
  - o Multiple-choice questions are worth 5 marks each.
  - o We may check your work, and if your answer is not justified by it, credit will not be given. You may use the back of the pages for rough work.
  - o Select the option that is numerically closest to your answer, circle it in this booklet, then copy it to the bubble sheet carefully using a pencil.
  - o **Only the bubble sheet will be marked.**
  - o If more than one choice is presented for a question (on either the test sheet or the bubble sheet), you may not get any credit for that question.
- **SECTION B: LONG-ANSWER QUESTIONS (#9—#12)**
  - o Long-answer questions are worth 10 marks each.
  - o For questions that require numerical answers, make sure you include all significant steps (such as equations and their numerical substitutions) in your solution, presented in the correct order. Unjustified answers will be given little or no credit. Cross out or erase all rough work not relevant to your solution. Put a box around your final answer.
  - o Write your solutions on the front of the page, below the questions. If you need more space, use the back of some page, but clearly indicate this fact on the original page, making sure that your solution cannot be confused with any rough work which may be there.

Please sign here [redacted] understood the above instructions.

Signature: [redacted]

DO NOT OPEN THE EXAM UNTIL YOU ARE TOLD TO DO SO

Then, before you start writing, verify that you **do** have all 10 pages (otherwise inform one of the invigilators).

$A = \frac{F_0}{(\omega_2-\omega_1)^2 - (\frac{b}{2m})^2}$

Section A: Multiple Choice Questions

1) One person swings on a swing and finds that the period is 3.0 s. Then a second person of equal mass joins him. With two people swinging, the period is

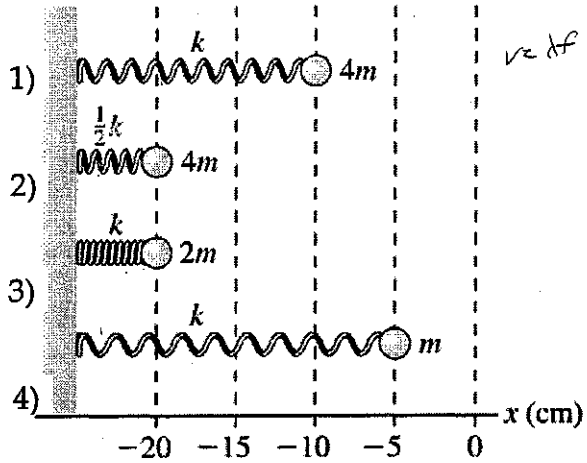
A. 1.5 s.
B. <3.0 s but not necessarily 1.5 s.
C. 3.0 s.
D. >3.0 s but not necessarily 6.0 s.
E. 6.0 s.

3.0s = T
m = m
2m = 2m
T =

Simple pendulum

mg sin θ would

2) Four springs have been compressed from their equilibrium position at x = 0 cm. When released, they will start to oscillate. Rank in order, from highest to lowest, the maximum speeds of the oscillators.



v = ∆f

A. $v_3 > v_2 > v_1 = v_4$
B. $v_3 > v_2 > v_1 > v_4$
C. $v_4 > v_1 > v_2 > v_3$
D. $v_1 = v_4 > v_2 > v_3$
E. $v_2 > v_3 > v_1 = v_4$

1) $\frac{k}{4m}$ 2) $\frac{k}{8m}$ 3) $\frac{k}{2m}$ 4) $\frac{k}{m}$

$\frac{4m}{k}$ $\frac{8m}{k}$ $\frac{2m}{k}$ $\frac{m}{k}$

3) Rank the waves represented by the following 3 functions according to their **period**, from the largest to the smallest.

$y1 = 2 \sin(3x + 10\,t + \pi)$ $\omega = \frac{2\pi}{T}$

$y2 = 8 \cos(6x + 19\,t)$

$y3 = 7 \sin(4x + 24\,t + 6)$

In the 5 options listed, T1 refers to the period of y1, etc.

Choose the correct option below:

A. T3>T2>T1

B. T1>T2>T3

C. T1>T3>T2

D. T2>T3>T1

E. T1>T3>T2

y1 $10 = \frac{2\pi}{T}$ $T = \frac{2\pi}{10}$

y2 $19 = \frac{2\pi}{T}$ $T = \frac{2\pi}{19}$

y3 $24 = \frac{2\pi}{T}$ $T = \frac{2\pi}{24}$

4) Which of the following actions would make a pulse travel **faster** down a stretched string?

A. Move your hand up and down **more quickly** as you generate the pulse.

B. Move your hand up and down **a larger distance** as you generate the pulse.

C. Use a **heavier** string of the same length, under the same tension.

D. Use a **lighter** string of the same length, under the same tension.

E. Use a **longer** string of the same thickness, density and tension.

Lighter string makes pulse faster due to the same amount

5) If the string does not stretch, by what factor would you have to multiply the tension in a taut string so as to double the wave speed?

A. 0.25

B. 4

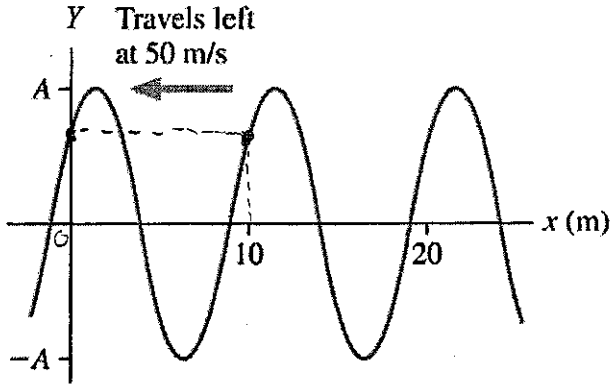
C. 2

D. 0.5

E. You could not change the speed by changing the tension.

Assuming wavelength remains constant

6) What is the frequency of this traveling wave (see figure)?



A. 10 Hz
B. 5 Hz
C. 2 Hz
D. 0.2 Hz
E. 0.1 Hz

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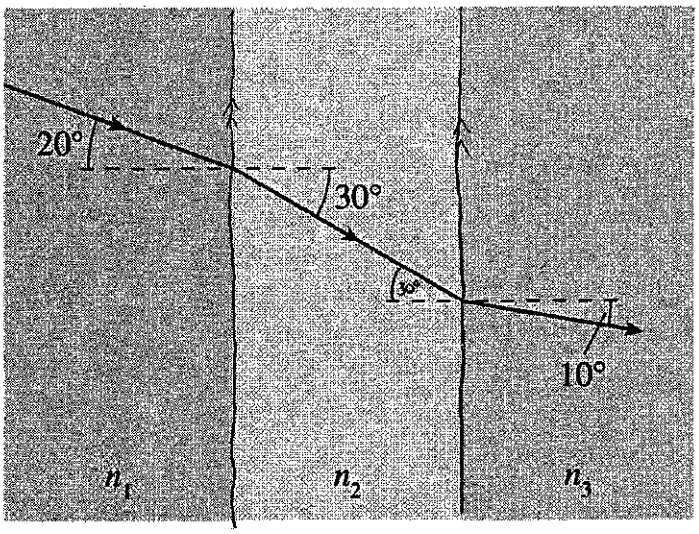
Amplitude = A
V = 50 m/s
λ = 10 m
T = 10 m

$v = \lambda f$
$50 = 10 f$
$f = \frac{50 m/s}{10 m}$
$f = 5 Hz$

7) A light ray travels from medium 1 to medium 2, then to medium 3 as shown. The dashed lines are normal to the interfaces.



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For these media, which relationship between indices of refraction is correct?

A. $n_3 < n_1$
B. $n_3 = n_1$
C. $n_3 > n_1$
D. We can't compare n_3 to n_1 without knowing n_2

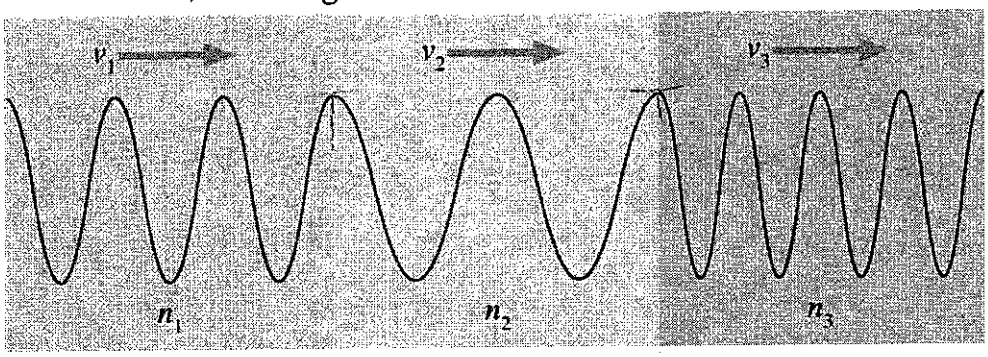
It is impossible to compare n_1 to n_3 if you don't have n_2

$n_1 \sin \theta_1 = n_2 \sin \theta_2$
$n_1 \sin 20 = n_2 \sin 30$
$\sin 30 > \sin 20$
$n_1 > n_2$

$\sin 30 < \sin 20$

$n_2 \sin \theta_2 = n_3 \sin \theta_3$
$n_2 \sin 30° = n_3 \sin 10°$
$\sin 30 > \sin 10$
$n_2 < n_1$

8) A light wave travels through three transparent materials of equal thickness. Rank in order, from largest to smallest, the indices of refraction n_1, n_2, and n_3.



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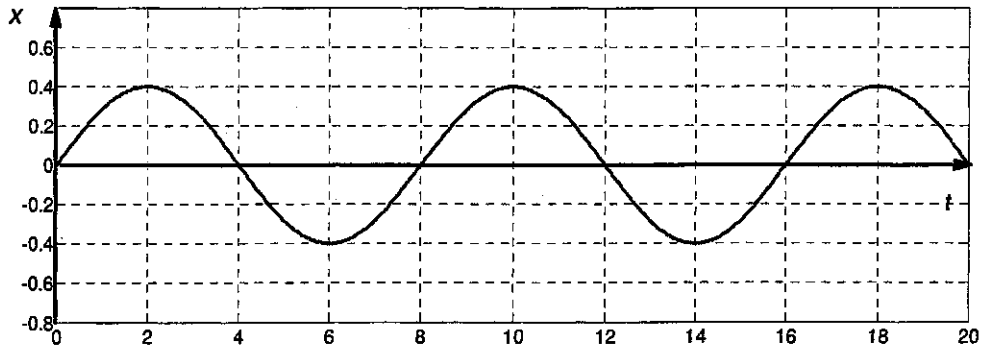
A. $n_2 > n_1 > n_3$
B. $n_3 > n_1 > n_2$
C. $n_1 > n_2 > n_3$
D. $n_3 > n_2 > n_1$
E. $n_2 = n_1 = n_3$

Section B: Long-Answer Questions

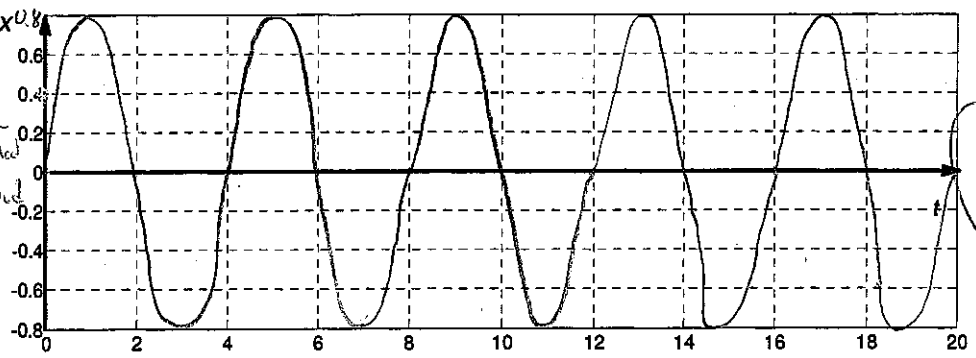
9) The graph below shows the position versus time for a mass oscillating on a spring, in some suitable units.



On the axes below sketch the position-versus-time plots for the following situations which differ in some respects from the original one. **Note: The changes described in each part refer back to the <u>original oscillation</u>, not to the oscillation of the previous part of the question.** Use the same horizontal and vertical scales as in the original graph.

a) The amplitude and frequency are doubled.

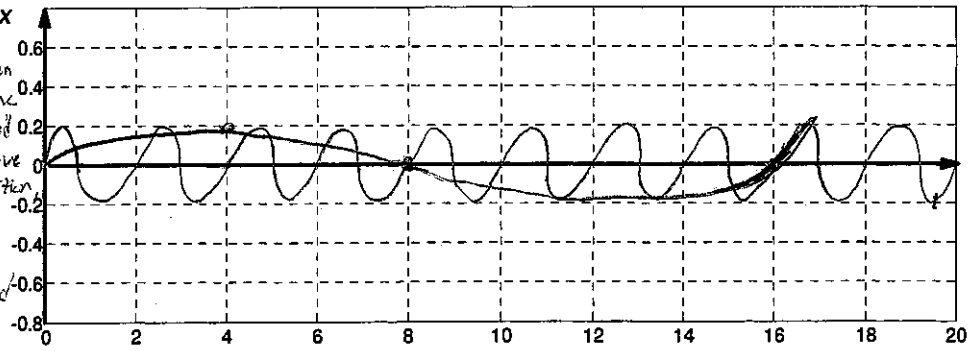
If amplitude is doubled then the position is doubled. If the frequency is doubled the time is halved



2

b) The amplitude is halved and the mass is quadrupled.

If amplitude is halved position is halved, but the mass is quadrupled so it should have double the position however if the amplitude alone is ~~doubled~~ halved perhaps mass would make the reactions go 4 times as fast

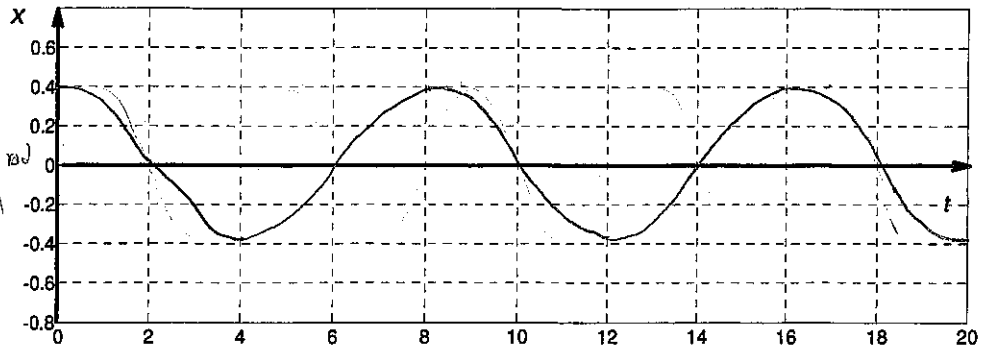


1

3

c) The phase constant is increased by π/2 rad. Phase constant = ∅

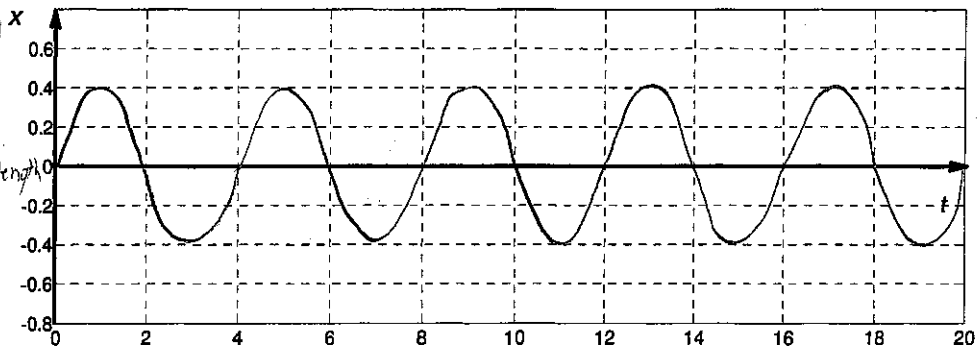
If the phase constant is increased by π/2 rad the graph will shift π/2 to the left



2

d) The maximum speed is doubled while the amplitude remains constant.

velocity is doubled meaning takes half the time to complete a wavelength

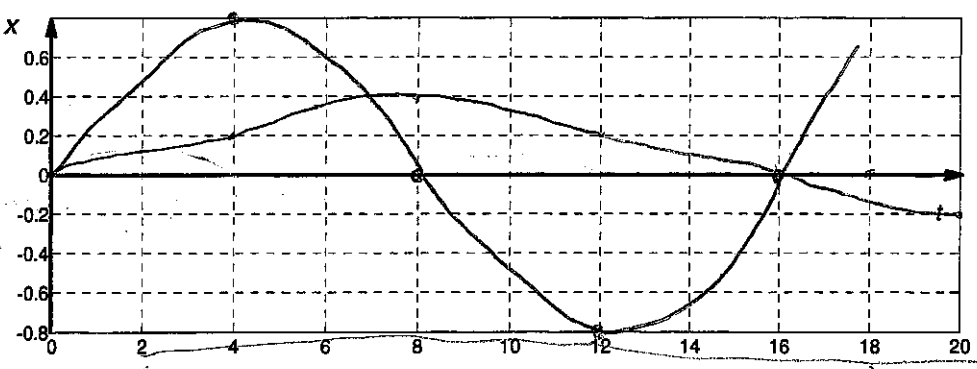


2

e) The maximum kinetic energy is quadrupled while the maximum speed remains constant.

$E_k = \frac{1}{2}mv^2$

If velocity ~~mass~~ remains constant and kinetic energy is quadruples that means that the mass must increase by 4.



0

If the mass is increased but the velocity stays the same then the amount of time it takes to complete a wavelength will be multiplied by 4, so t = 4x original t. v = m/s If v stays the same and m is 4x then s has to be 4x.

$E_k = \frac{1}{2}mv^2$

$E_k = 0.5mv^2$

$4E_k = \frac{1}{2}4mv^2$

$4E_k = 2mv^2$

$v_1 = m/s$ $v_2 = 4m/4s$

$v_1 = v_2$

$\therefore v_2 = 4m/4s = m/s$

10) Damping is negligible for a 0.100 kg object hanging from a light spring with spring constant 6.40 N/m. A sinusoidal force with amplitude of 1.00 N drives the system. At what angular frequencies will the force make the object vibrate with an amplitude of 0.250 m?

Hint: $A = \dfrac{F_0/m}{\sqrt{(\omega^2-\omega_0^2)^2+\left(\dfrac{b}{2m}\right)^2}}$

$m = 0.1$ kg $\quad \omega_0 = ?$

$b = 6.40$ N/m $\quad \omega = ?$

$F_0 = 1.00$ N (1)

$A = 0.250$ m (1)

$$A = \frac{F_0/m}{\sqrt{(\omega^2-\omega_0^2)^2+\left(\frac{b}{2m}\right)^2}}$$

$$A^2 = \frac{F_0^2/m^2}{(\omega^2-\omega_0^2)^2+\left(\frac{b}{2m}\right)^2}$$

$$\frac{F_0^2}{m^2A^2} = (\omega^2-\omega_0^2)^2+\left(\frac{b}{2m}\right)^2 \quad (1)$$

$$\frac{F_0^2}{m^2A^2} - \left(\frac{b}{2m}\right)^2 = (\omega^2-\omega_0^2)^2$$

$$\frac{1^2}{(0.1)^2(0.25)^2} - \left(\frac{6.4}{2(0.1)}\right)^2 = (\omega^2-\omega_0^2)^2$$

$$1600 - 1024 = (\omega^2-\omega_0^2)^2$$

$$576 = (\omega^2-\omega_0^2)^2$$

$$24 = \omega^2-\omega_0^2$$

Assume $\omega_0 = 0$

$$24 = \omega^2$$

$$\omega = \sqrt{24}$$

$$\omega = 2\sqrt{6} \text{ rad/s}$$

$\therefore$ Both angular frequencies when subtracted from their squares and equal 24 or if ω_0 is equal to 0 then ω is $2\sqrt{6}$.

Given $A = 0.250$ m

$b \approx 0$

$F_0 = 1.00$ N

$m = 0.1$ kg

$k = 6.40$ N/m

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{6.40}{0.100}} = 8.00 \text{ rad/s}$$

If $b \approx 0$, $\therefore A = \dfrac{F_0/m}{\sqrt{(\omega^2-\omega_0^2)^2}} = \dfrac{F_0/m}{\pm(\omega^2-\omega_0^2)}$

$$\therefore \omega^2 - \omega_0^2 = \pm\frac{F_0}{Am}$$

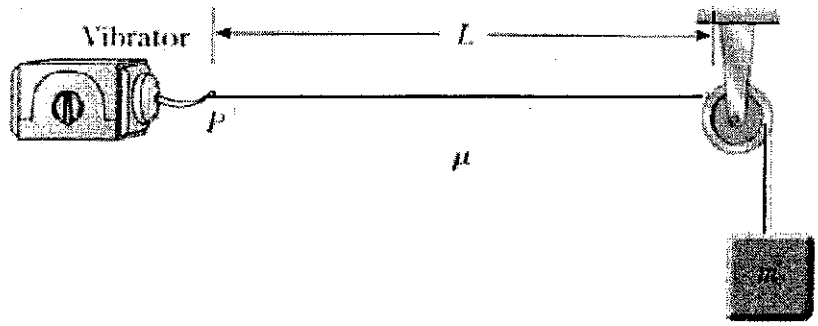
By definition, $\omega > 0$, $\therefore \omega = \sqrt{\omega_0^2 \pm \frac{F_0}{Am}}$

$$\therefore \omega = \sqrt{(8.00)^2 \pm \frac{1.00}{(0.250)(0.100)}} = \sqrt{64 \pm 40} \text{ rad/s}$$

i.e., $\omega_1 = \sqrt{104}$ rad/s $= 10.19$ rad/s

or $\omega_2 = \sqrt{24}$ rad/s $= 4.90$ rad/s

11) In the arrangement shown an object is hung from a string (with a linear mass density of 0.00200 kg/m) that passes over a light pulley. The string is connected to a vibrator of frequency f=350Hz. The length of the string between point P and the pulley is L=2.00 m. What is the largest mass, *m*, for which a standing wave can be observed?



$\mu = 0.00200\ kg/m$

$f = 350\ Hz$

$L = 2.00\ m$

$m = ?$

$= \sqrt{\frac{L}{\mu}}$

$\omega = \sqrt{\frac{g}{L}}$

$\omega = \sqrt{\frac{9.8}{2}}$

$\omega = \sqrt{4.9}$

$\omega = 2.213594362$

$\omega \doteq 2.21\ rad/s$

$m = \sqrt{\mu \times L}$

$m = \sqrt{0.00200\ kg/m \times 2.00\ m}$

$m = \sqrt{0.00400\ kg}$

$m = 0.063245553\ kg$

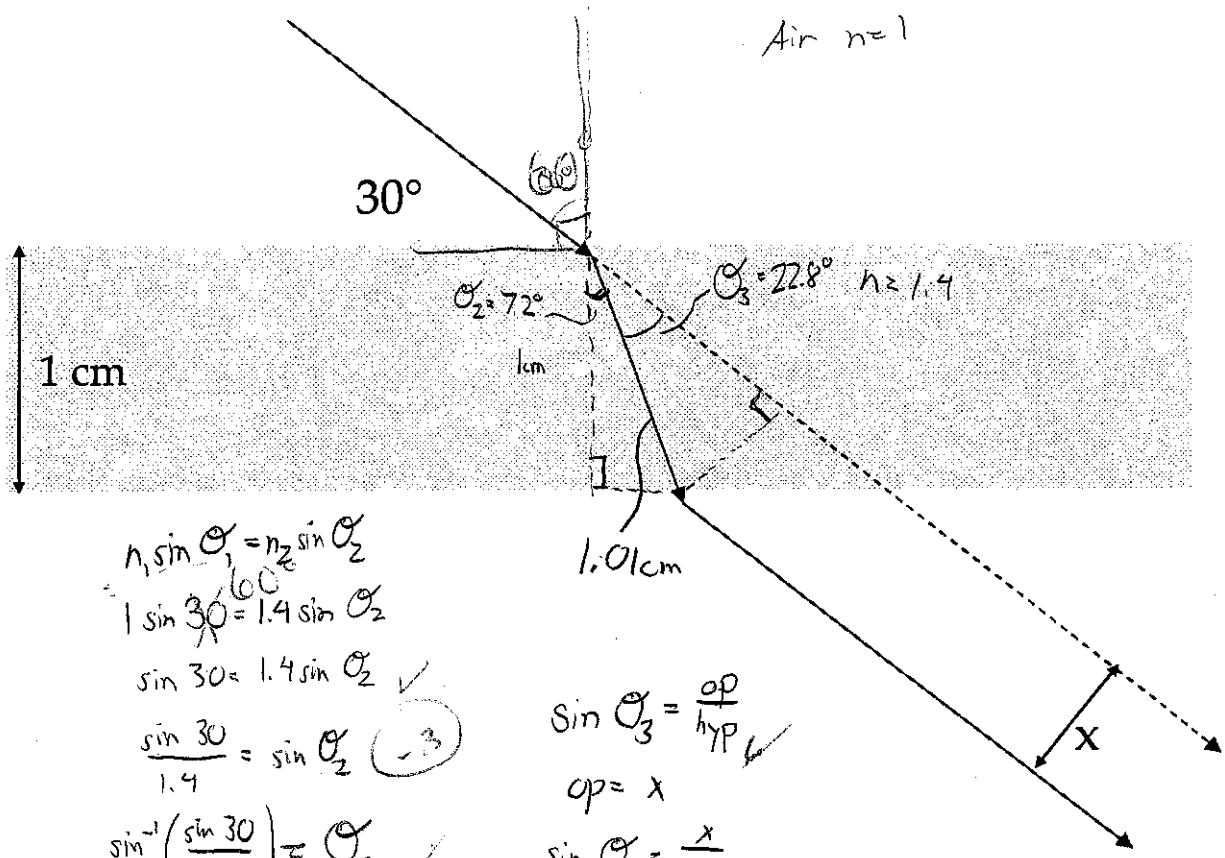
$\therefore m \doteq 63.2\ g$

Largest mass for which a standing wave can be observed

zero

0

12) Find the lateral displacement X (see figure) of the light beam passing through a planar transparent plate of thickness 1 cm and index of refraction 1.4, surrounded by air.



$n_1 \sin\theta_1 = n_2 \sin\theta_2$

$1 \sin 30 = 1.4 \sin\theta_2$ (60)

$\sin 30 = 1.4 \sin\theta_2$

$\frac{\sin 30}{1.4} = \sin\theta_2$ (-3)

$\sin^{-1}\left(\frac{\sin 30}{4}\right) = \theta_2$

$\theta_2 = 7.18075782°$

$\theta_2 \doteq 7.2°$ X

$\theta_3 = 30° - 7.2°$

$= 22.8°$

$\sin\theta_3 = \frac{op}{hyp}$

$op = x$

$\sin\theta_3 = \frac{x}{hyp}$

$\sin 22.8° = \frac{x}{1.01cm}$

$1.01cm \sin 22.8° = x$

$x = 0.391390742 cm$

$\boxed{\therefore x \doteq 0.39 cm}$

(-1)

$\cos\theta = \frac{adj}{hypotenuse}$

$\frac{adj}{\cos\theta} = hypotenuse$

$hypotenuse = \frac{1cm}{\cos 7.2}$

$hypotenuse = 1.00794797 cm$

$\doteq 1.01 cm$

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